



Home Automation

using Arduino & Bluetooth

Smart Home | Manual and Smartphone Control



What We Will Cover

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Introduction & Problem Statement

Why smart homes?

02

What is Arduino?

The brain of our project

03

Bluetooth Module (HC-05)

Wireless communication

04

Circuit Design

Connections & components

05

Mobile App Control

Smartphone operation

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Working Demo

Manual + App modes

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Applications & Future Scope

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Key takeaways

Why Home Automation?

Convenience



Control lights, fans & appliances from your phone — no need to walk to every switch.

Energy Saving



Turn off forgotten devices automatically — save electricity & reduce bills.

Security



Manage home devices remotely even when you're away from home.

Accessibility



Helps elderly and differently-abled people operate home appliances easily.

Arduino Uno — The Brain

- Open-source microcontroller board
- ATmega328P chip — 16 MHz clock speed
- 14 Digital I/O pins | 6 Analog pins
- Operates at 5V, easily powered via USB
- Programmable using Arduino IDE (C/C++)
- Controls relays, sensors, motors & more

Specification	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Digital I/O Pins	14 (6 PWM)
Analog Pins	6
Flash Memory	32 KB
Clock Speed	16 MHz
USB Interface	Type-B

⚡ In our project, Arduino Uno reads Bluetooth commands and switches home appliances ON/OFF via relay modules.

HC-05 Bluetooth Module



Wireless Range

Up to 10 metres
Indoor line-of-sight



Communication

UART Serial
Baud: 9600 bps



Pairing

Master/Slave mode
Pairs with smartphone

HC-05 Pin	Connect To	Purpose
VCC	5V (Arduino)	Power supply
GND	GND (Arduino)	Ground
TXD	Pin 10 (RX)	Transmit data
RXD	Pin 11 (TX)	Receive data

Components & Circuit | 04

Components Required



Arduino Uno

1 unit



HC-05 Bluetooth

1 unit



4-Channel Relay

1 unit



LED / Bulb

4 units



9V Battery / Adapter

1 unit



Jumper Wires

As needed



Android Smartphone

1 unit



Breadboard

1 unit

Circuit Flow Diagram



Smartphone App



HC-05 Bluetooth



Arduino Uno

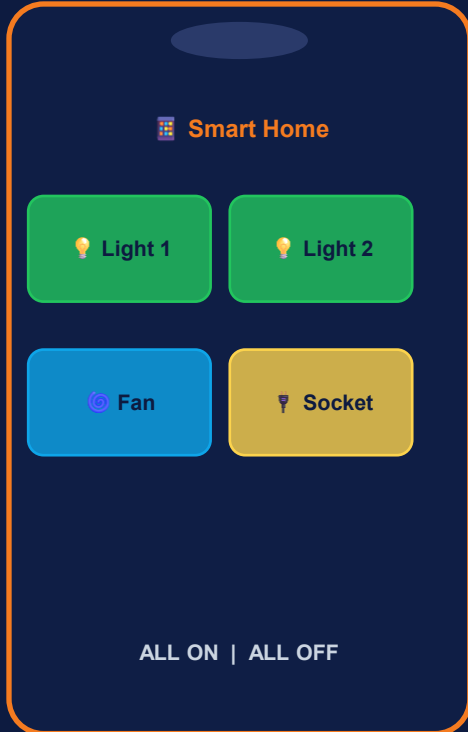


Relay Module



Appliances

Smartphone Control via Bluetooth



1

Download App

Install 'Arduino Bluetooth Controller' or 'BT Voice Control' from Play Store

2

Pair Bluetooth

Enable Bluetooth on phone → Pair with HC-05 (Password: 1234 or 0000)

3

Connect & Open

Launch the app → Connect to HC-05 module

4

Assign Commands

Map buttons: '1' = Light ON, '0' = Light OFF, '2' = Fan ON, etc.

5

Control!

Tap buttons to toggle appliances wirelessly from smartphone

Two Modes of Operation



Manual Mode

- Physical push buttons on control panel
- Directly wired to Arduino digital pins
- Arduino reads HIGH/LOW signal
- Relay switches the appliance circuit
- LED indicator shows current state
- Works even without Bluetooth/phone

VS



Smartphone Mode

- Open Bluetooth controller app on phone
- Connect to HC-05 module (up to 10m)
- Send character command (e.g. '1', '0')
- HC-05 forwards to Arduino via serial
- Arduino activates/deactivates relay
- Appliance turns ON or OFF instantly

Sample Arduino Sketch

```
// Home Automation - Arduino Bluetooth Code
#define RELAY1 8 // Light
#define RELAY2 9 // Fan

void setup() {
  Serial.begin(9600);
  pinMode(RELAY1, OUTPUT);
  pinMode(RELAY2, OUTPUT);
}

void loop() {
  if (Serial.available()) {
    char cmd = Serial.read();
    if (cmd == '1') digitalWrite(RELAY1, LOW); // Light ON
    if (cmd == '0') digitalWrite(RELAY1, HIGH); // Light OFF
    if (cmd == '2') digitalWrite(RELAY2, LOW); // Fan ON
    if (cmd == '3') digitalWrite(RELAY2, HIGH); // Fan OFF
  }
}
```

Real-World Applications



Smart Home

Control lights, fans, AC & TV automatically



Offices

Centralized building automation



Hospitals

Remote control for patient rooms



Disabled Support

Voice/phone control for mobility aid



Industries

Automated machinery control

Future Enhancements



Wi-Fi / IoT Control

Control from anywhere using internet



Voice Commands

Add Google Assistant / Alexa support



Energy Monitor

Track electricity usage in real-time



Smart Security

Add motion sensors & camera alerts



AI Automation

Auto-schedule based on usage patterns

Key Takeaways



Simple & Affordable

Built with low-cost, easily available components



Dual Control

Works both manually and via smartphone Bluetooth



Practical Learning

Covers electronics, coding, and IoT concepts



Energy Efficient

Reduces unnecessary power consumption



Expandable

Can be extended to full smart home with Wi-Fi/IoT



Thank You!

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Questions are Welcome!

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